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Crystal arthropathies

Deconstruction of tophi and synovium defines SPP1⁺ macrophages involved in extracellular matrix remodelling in gout

Hanlin Xu[1,****], Zhechen Liu[1,2], Xiaofeng Zhou[1,2], Xiaoxi Ji[1], Xingwang Liu[1], Xiaoxia Zhu[3], Liwei Lu[4], Nicola Dalbeth[5,***], Rui He[6,7,**], Yinghui Hua[1,*]

[1] Department of Sports Medicine, Huashan Hospital, Fudan University, Shanghai, China
[2] Clinical Medical College, Fudan University, Shanghai, China
[3] Department of Rheumatology, Huashan Hospital, Fudan University, Shanghai, China
[4] Department of Pathology and Shenzhen Institute of Research and Innovation, The University of Hong Kong, Hong Kong, China
[5] Department of Medicine, The University of Auckland, Auckland, New Zealand
[6] Department of Immunology, School of Basic Medical Sciences, Fudan University, Shanghai, China
[7] National Clinical Research Center for Aging and Medicine, Huashan Hospital, Fudan University, Shanghai, China

A R T I C L E I N F O

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A B S T R A C T

Objectives: Gout is a prevalent condition characterised by high serum urate levels, leading to crystal deposition in the joints. Although many studies have explored the mechanisms underlying gout flares, the dynamics of tophus formation remain unknown. This study provides the first transcriptomic profile of the tophaceous gout joint and investigates immune-stromal cell interactions in the pathogenesis of tophus formation.
Methods: Single-cell transcriptomic profiling was conducted on 44,221 synovial tissue cells from patients with intercritical gout (without tophi) and tophaceous gout. Spatial transcriptomics showed gene expression patterns in the corona and fibrovascular zones of tophi. Gene expression pattern comparisons, pseudotime, and differential gene enrichment analyses were conducted on stage-specific macrophage subsets. Immunofluorescence and flow cytometry on the tophi samples validated the transcription results and visualised the spatial localisation of the macrophage-fibroblast subpopulation. Differential causal inference combined with Mendelian randomisation elucidated gene regulatory networks and their causal relationships with gout pathology.
Results: We identified *SPP1⁺/MMP9⁺/CHI3L1⁺* macrophages within the corona zone exclusive to tophaceous gout, exhibiting extracellular matrix regulation genes, enhanced integrin-mediated interactions with stromal cells and transitional potential towards osteoclast differentiation. Notably, coexpression of the fibroblast marker *S100A4* and *COL6A2* in these macrophages suggested a fibroblast-like phenotype. Distinct *CD4⁺* T-cell transcriptional profiles between disease states indicated a phenotypic shift from inflammatory to immune-regulatory subsets during tophus development.

***Correspondence to** Prof. Yinghui Hua, Department of Sports Medicine, Huashan Hospital, Fudan University, Shanghai, China.
****Correspondence to** Prof. Rui He.
*****Correspondence to** Prof. Nicola Dalbeth.
******Correspondence to** Dr. Hanlin Xu.
E-mail addresses: hanlinxu@fudan.edu.cn (H. Xu), n.dalbeth@auckland.ac.nz (N. Dalbeth), ruihe@fudan.edu.cn (R. He), hua_cosm@aliyun.com (Y. Hua).
Handling editor Josef S. Smolen.
HX, ZL and XZ contributed equally.

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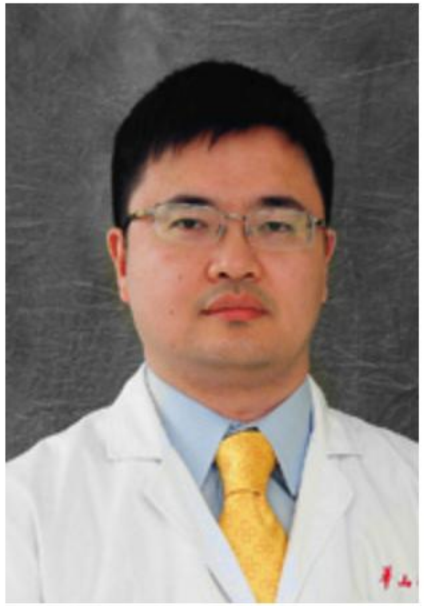
XIANGYA HOSPITAL CENTRAL SOUTH UNIVERSITY

journal club presentation

刘志

(BGI Qingdao C-LiuZhi)

2 June 2026



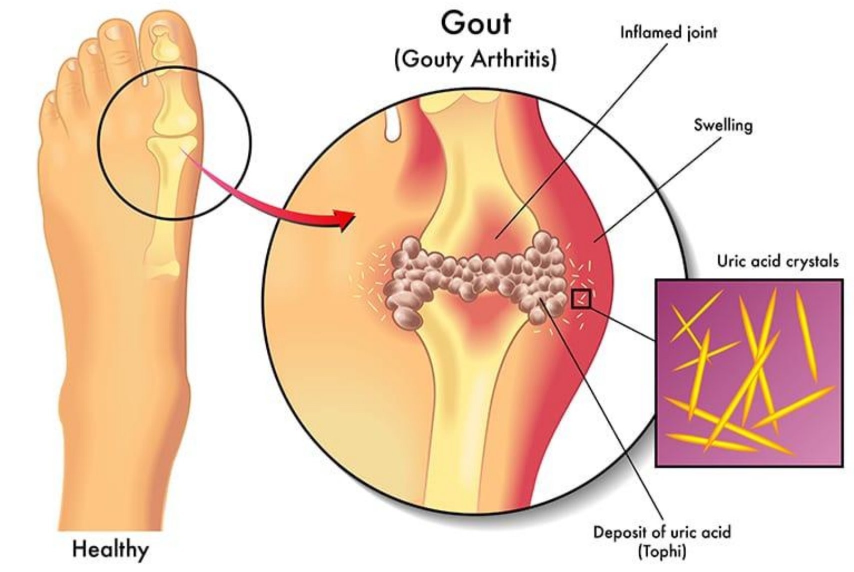
1 [Deconstruction of tophi and synovium defines SPP1(+) macrophages involved in extracellular matrix remodelling in **gout**.](#)
Cite
Ann Rheum Dis | 20.6 | Q1 | 2025 Dec
Xu H, Liu Z, Zhou X, Ji X, Liu X, Zhu X, Lu L, Dalbeth N, He R, **Hua Y.**
DOI: 10.1016/j.ard.2025.09.003
PMID: 41107120
OBJECTIVES: **Gout** is a prevalent condition characterised by high serum urate levels, leading to crystal deposition in the joints. ...METHODS: Single-cell transcriptomic profiling was conducted on 44,221 synovial tissue cells from patients with intercritical **gout** (wit ...
Abstract | Search in PubMed | Times Cited: 0 | Heat: 743 | Tags:

4 Contributions of joint damage-related events to **gout** pathogenesis: new insights from laboratory research.
Cite
Ann Rheum Dis | 20.6 | Q1 | 2023 Dec | Review
Xu H, Qin H, **Hua Y**, Dalbeth N.
DOI: 10.1136/ard-2023-224679
PMID: 37586760
Epidemiological and imaging findings indicate that **gout** frequently affects damaged joints. Recent studies suggest that the relationship between **gout** and joint damage may be more complex than a simple unidirectional link and that joint damage may promote the developm ...
Abstract | Search in PubMed | Times Cited: 4 | Heat: 372

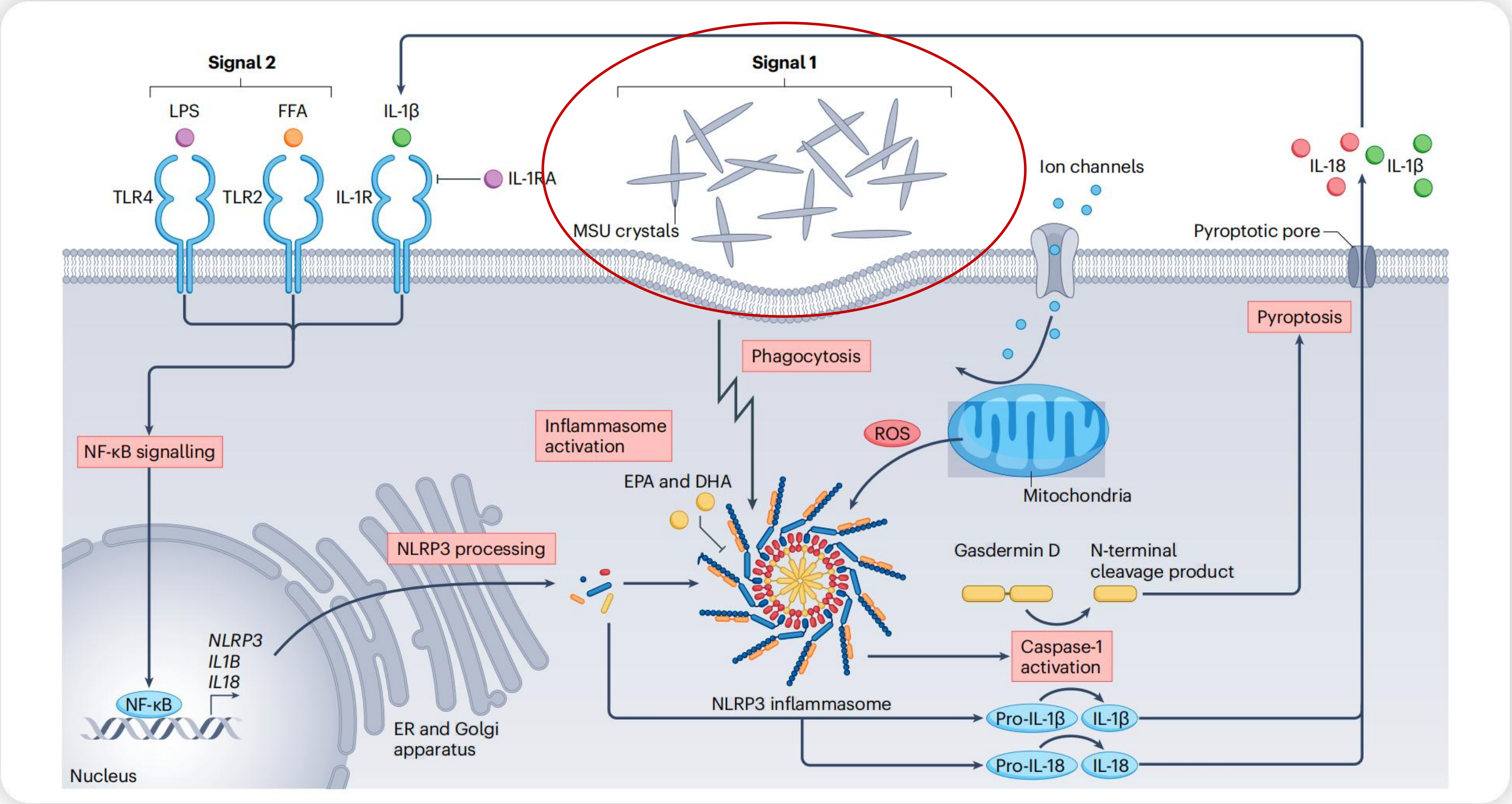
7 Type II collagen facilitates gouty arthritis by regulating MSU crystallisation and inflammatory cell recruitment.
Cite
Ann Rheum Dis | 20.6 | Q1 | 2023 Mar
Xu H, Zhang B, Chen Y, Zeng F, Wang W, Chen Z, Cao L, Shi J, Chen J, Zhu X, Xue Y, He R, Ji M, **Hua Y.**
DOI: 10.1136/ard-2022-222764
PMID: 36109143
OBJECTIVE: Increasing evidence suggests that impaired cartilage is a substantial risk factor for the progression from hyperuricaemia to **gout**. Since the relationship between cartilage matrix protein and **gout** flares remains unclear, we investigated its role in monosod ...
Abstract | Search in PubMed | Times Cited: 26 | Heat: 1513

3 Spatiotemporal Observation of Monosodium Urate Crystals Deposition in Synovial Organoids Using Label-Free Stimulated Raman Scattering.
Cite
Research (Wash D C) | 10.9 | Q1 | 2024 May 27
Chen Y, Chen Z, Wang W, **Hua Y**, Ji M.
DOI: 10.34133/research.0373
PMID: 38803506
Gout, a common form of arthritis, is characterized by the deposition of monosodium urate (MSU) crystals in joints. ...Our work contributes to the fundamental understanding of MSU-based diseases and offers valuable insights for the future development of targeted **gout** ...
Abstract | Search in PubMed | Times Cited: 4 | Heat: 134

2023	2023	2024	2025
Ann Rheum Dis	**Ann Rheum Dis**	**Research**	**Ann Rheum Dis**
机制实验	理论框架	成像模型	单细胞/空间组学
CII facilitates gouty arthritis	**Joint damage-related events**	**Spatiotemporal MSU deposition**	**$SPP1^+$ macrophages in tophi**
软骨 ECM 成分影响 MSU 结晶，并增强炎症细胞募集	提出“受损关节局部微环境”参与痛风发生与部位偏好	用 SRS + 滑膜类器官动态观察 MSU crystals 沉积过程	发现 $SPP1^+$ TGAMs 连接慢性炎症、ECM remodeling 和痛风石形成
PMID 36109143	PMID 37586760	PMID 38803506	PMID 41107120



urate transporter：SLC2A9、SLC22A12 and ABCG2

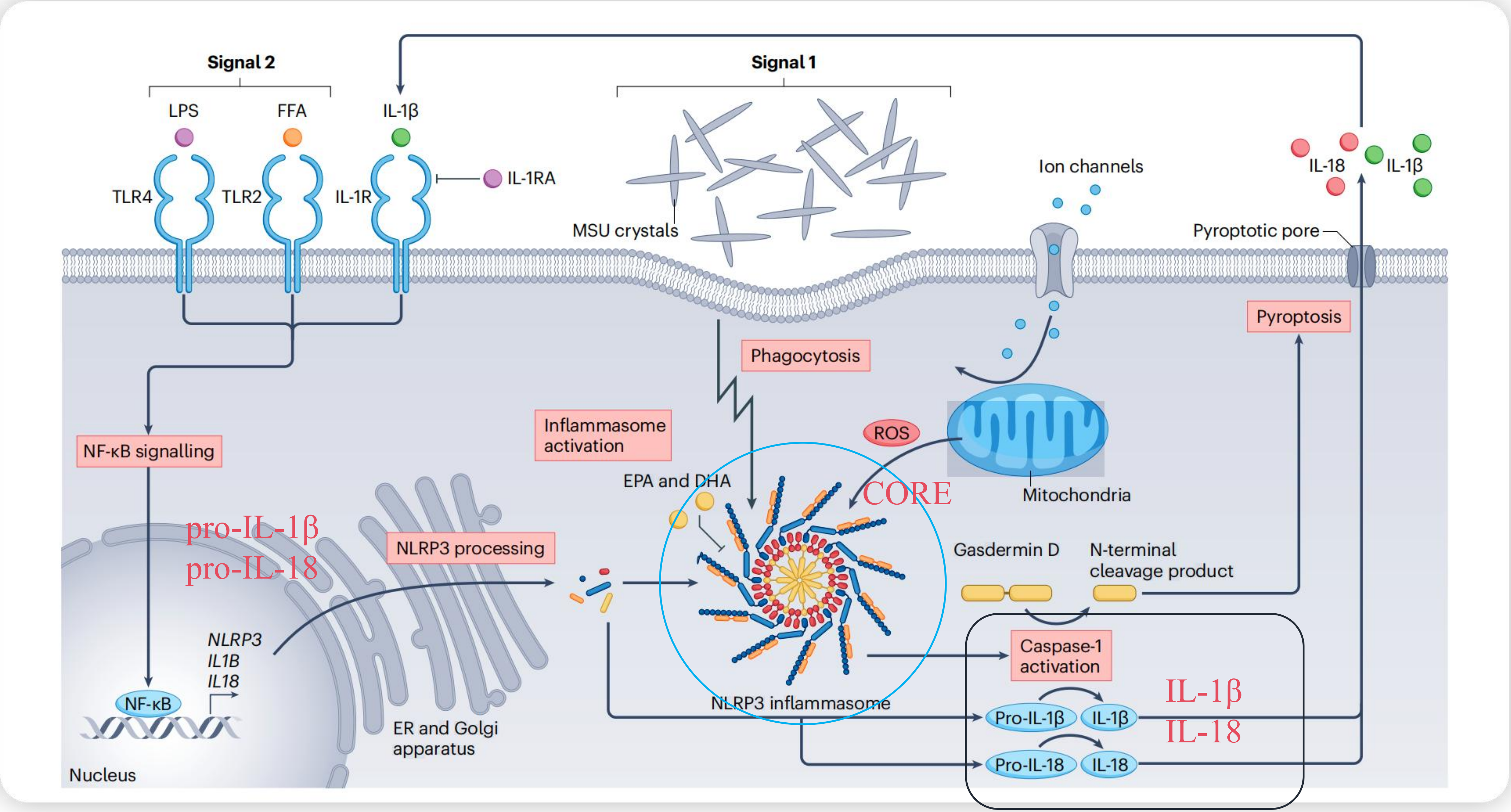


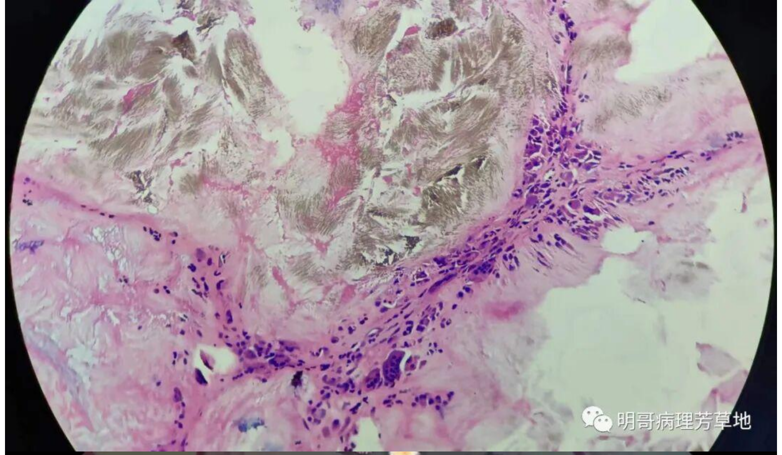
单钠尿酸盐晶体 monosodium urate crystals, MSU crystals

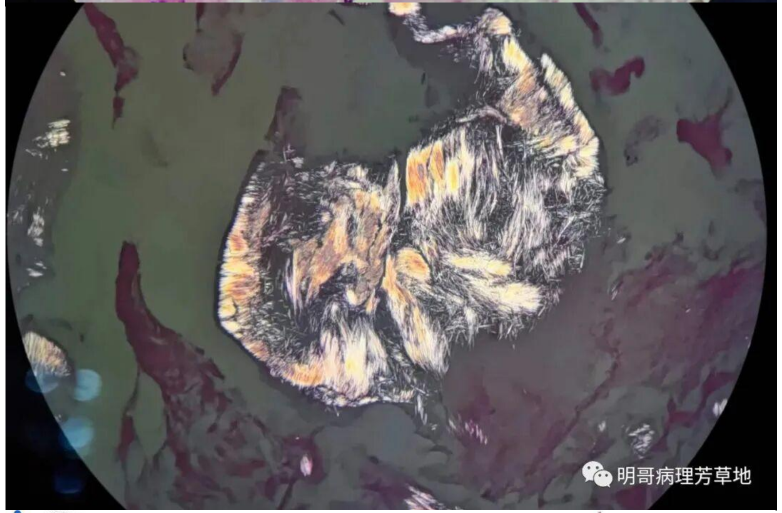
1.Dalbeth N, et al. Gout. Lancet. 2021;397:1843-1855.
2.Leask MP, et al. The pathogenesis of gout: molecular insights from genetic, epigenomic and transcriptomic studies. Nat Rev Rheumatol. 2024;20:510-523.
3.FitzGerald JD, et al. 2020 American College of Rheumatology Guideline for the Management of Gout. Arthritis Care Res. 2020;72:744-760.

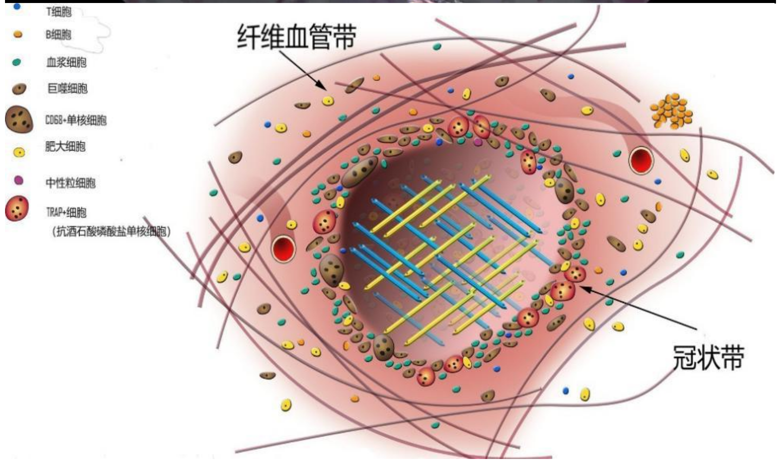
MSU crystals → macrophage phagocytosis → NLRP3 inflammasome activation → caspase-1 → IL-1β release and pyroptosis → acute gout flare

MSU 晶体通过激活巨噬细胞的 NLRP3 inflammasome，引发 IL-1β 释放和细胞焦亡，从而驱动急性痛风发作

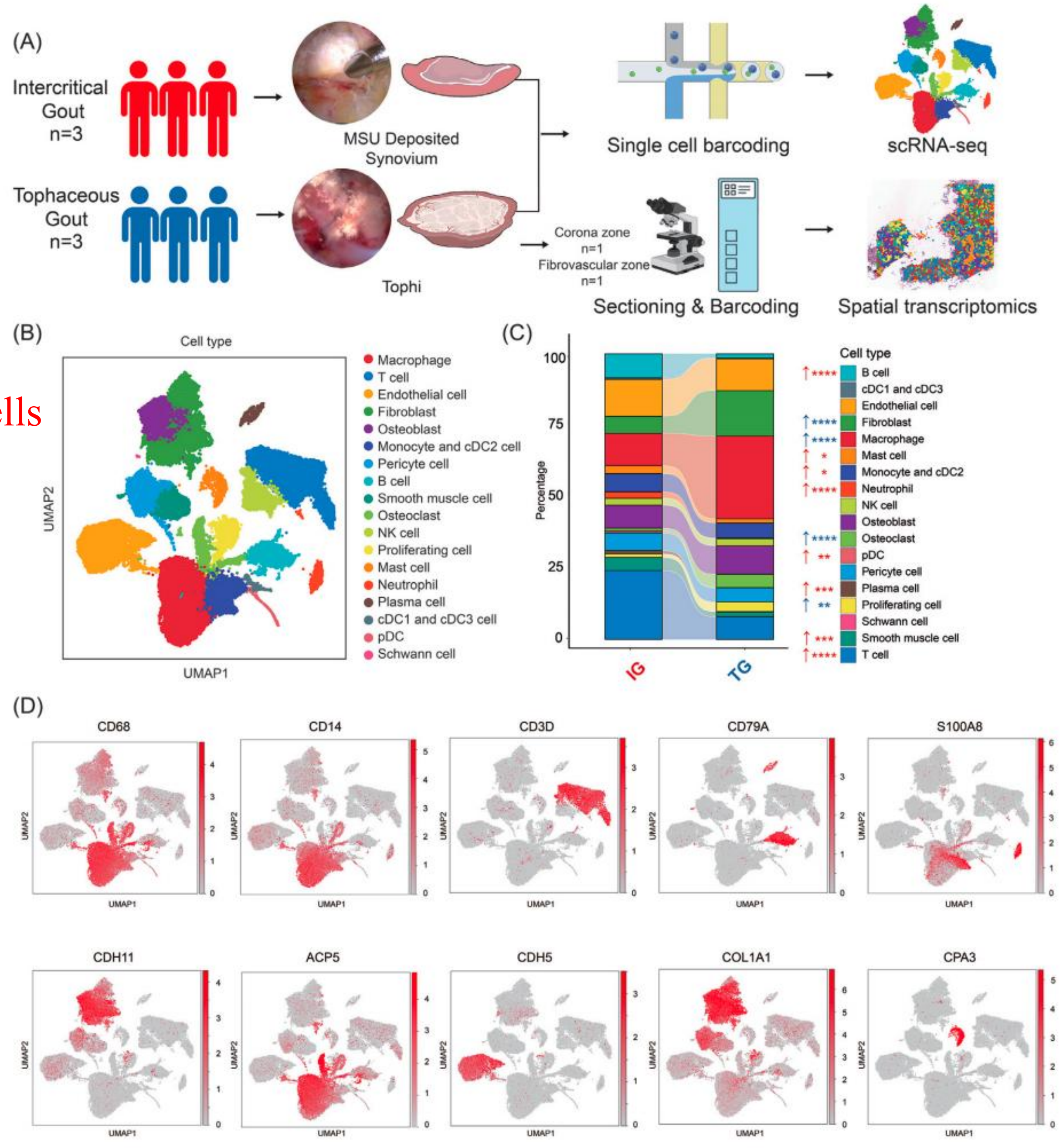


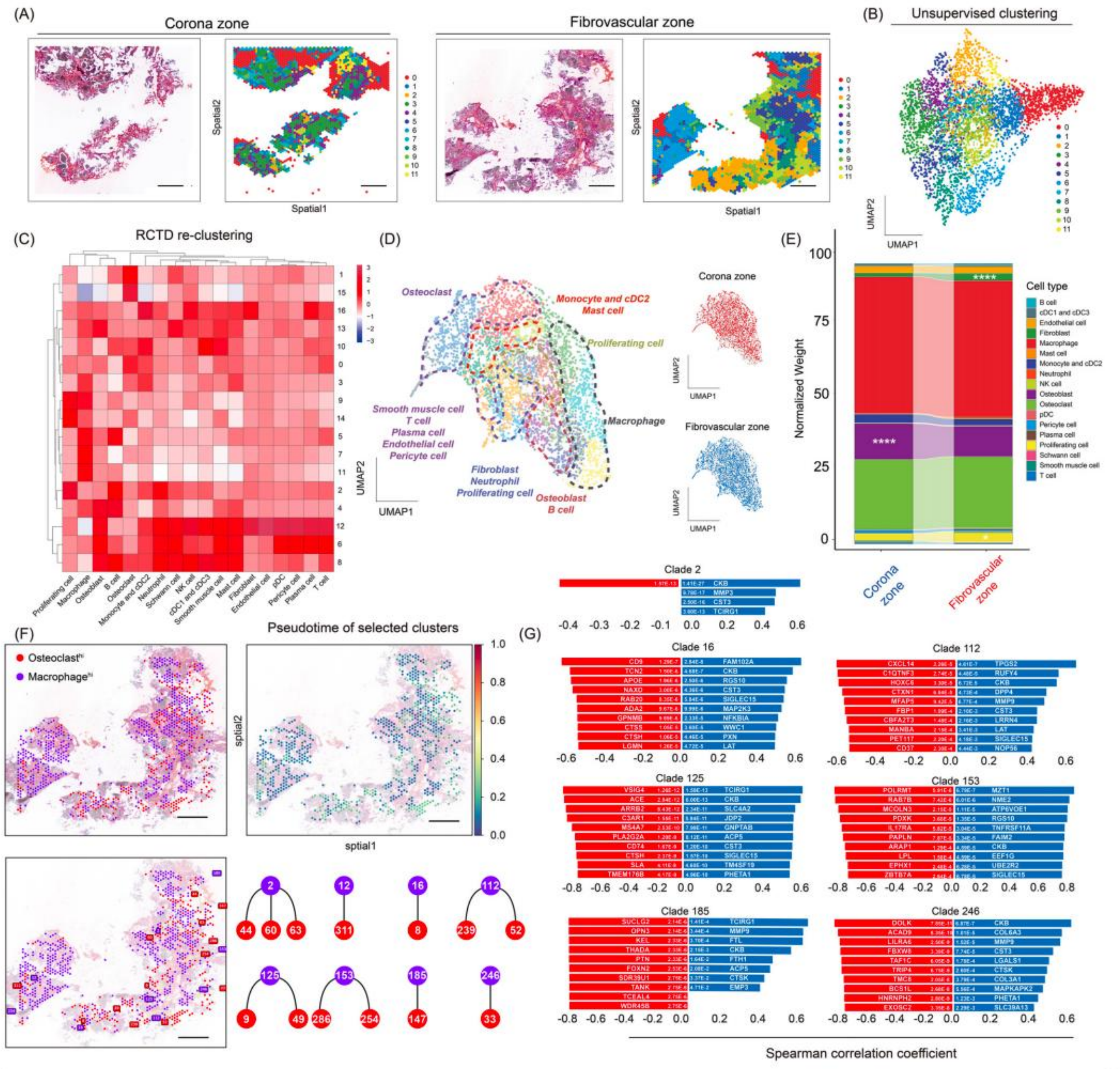

明哥病理芳草地


明哥病理芳草地

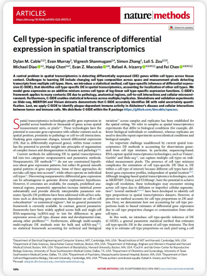

T细胞
B细胞
血浆细胞
巨噬细胞
CD68+单核细胞
肥大细胞
中性粒细胞
TRAP+细胞
(抗酒石酸磷酸盐单核细胞)
纤维血管带
冠状带

44,221Cells


(A)
Intercritical Gout n=3
Tophaceous Gout n=3
MSU Deposited Synovium
Tophi
Single cell barcoding
scRNA-seq
Corona zone n=1
Fibrovascular zone n=1
Sectioning & Barcoding
Spatial transcriptomics
(B)
Cell type
UMAP1
UMAP2
Macrophage
T cell
Endothelial cell
Fibroblast
Osteoblast
Monocyte and cDC2 cell
Pericyte cell
B cell
Smooth muscle cell
Osteoclast
NK cell
Proliferating cell
Mast cell
Neutrophil
Plasma cell
cDC1 and cDC3 cell
pDC
Schwann cell
(C)
Percentage
IG
TG
Cell type
B cell
cDC1 and cDC3
Endothelial cell
Fibroblast
Macrophage
Mast cell
Monocyte and cDC2
Neutrophil
NK cell
Osteoblast
Osteoclast
pDC
Pericyte cell
Plasma cell
Proliferating cell
Schwann cell
Smooth muscle cell
T cell
(D)
CD68
CD14
CD3D
CD79A
S100A8
CDH11
ACP5
CDH5
COL1A1
CPA3

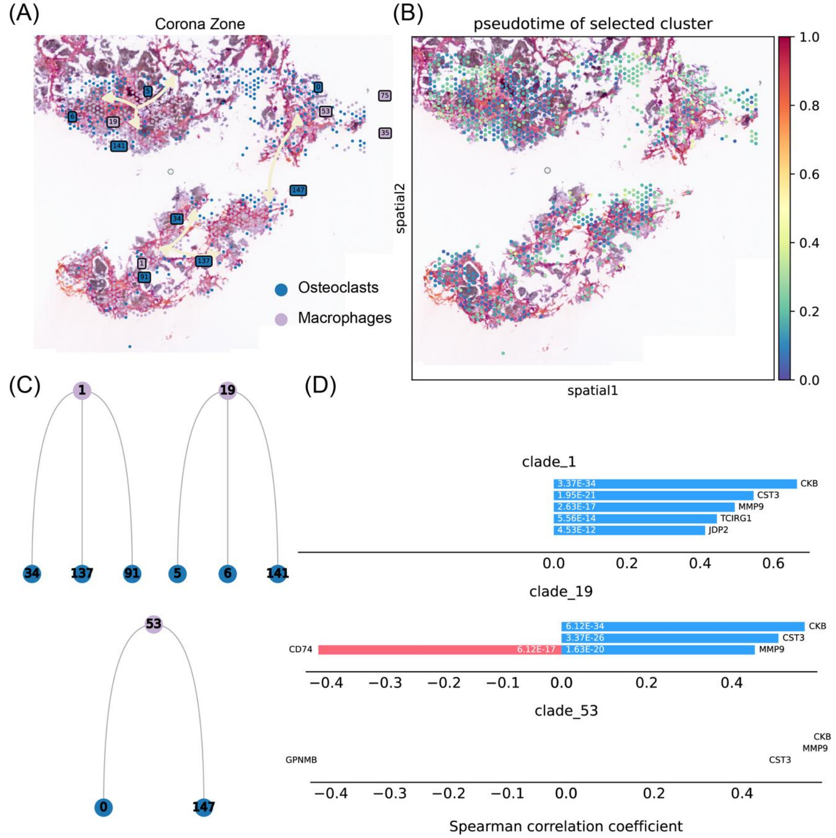


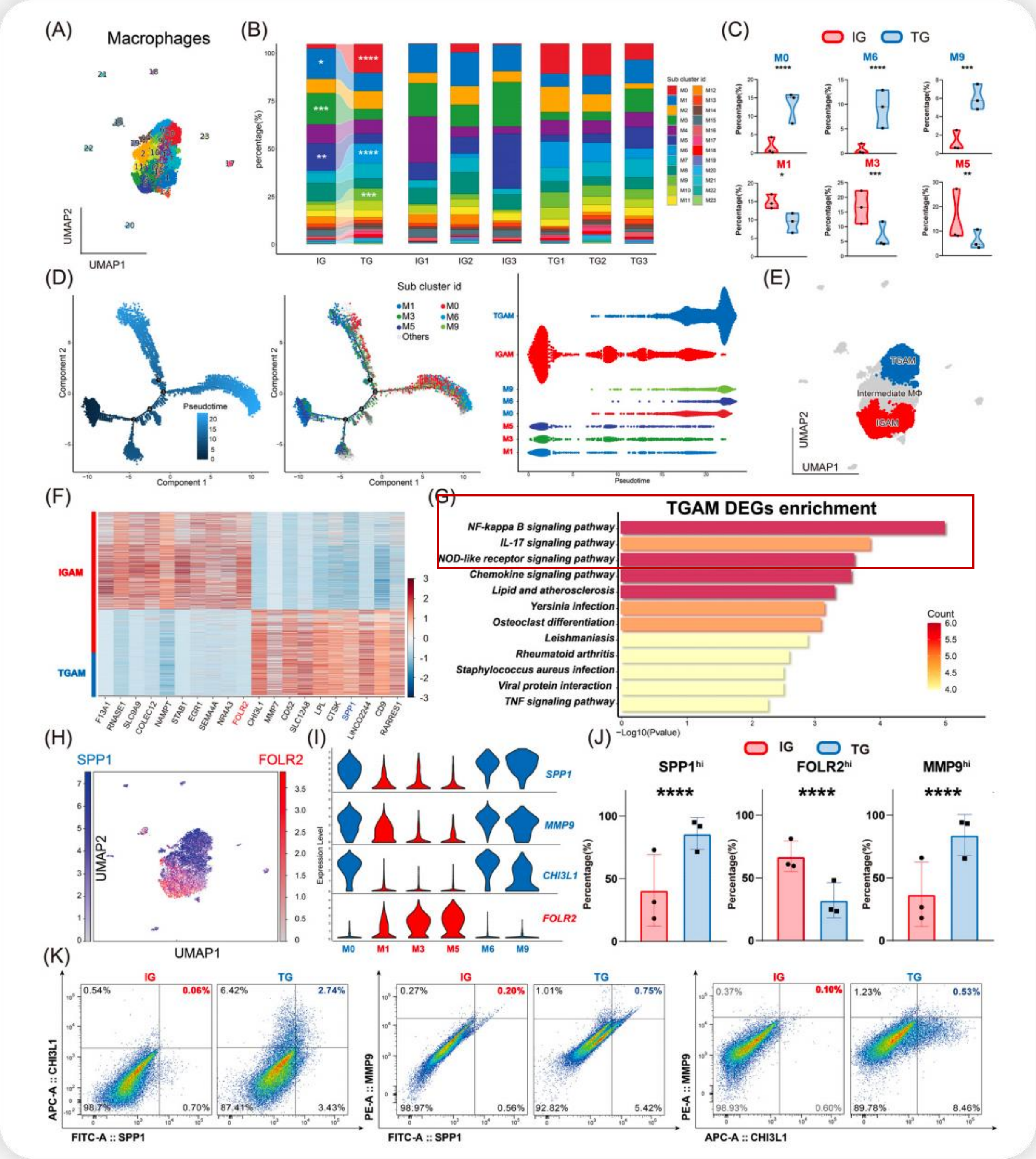
ST
LEARN

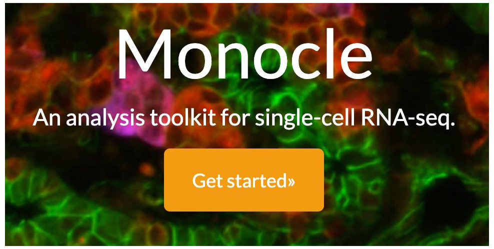


https://github.com/BiomedicalMachineLearning/stLearn

https://github.com/dmcable/spacexr







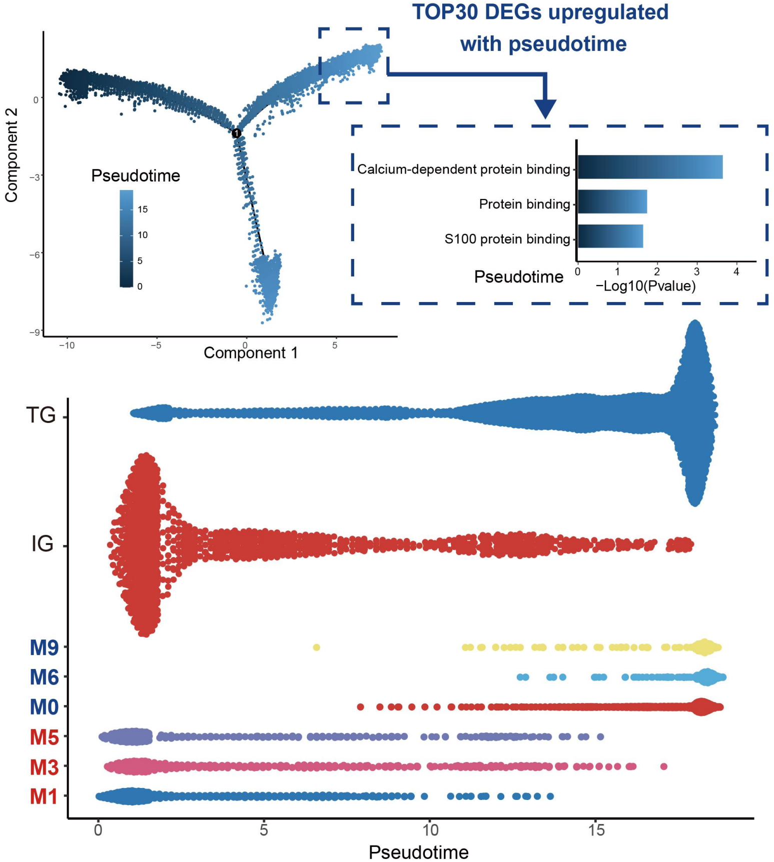
https://cole-trapnell-lab.github.io/monocle-release/

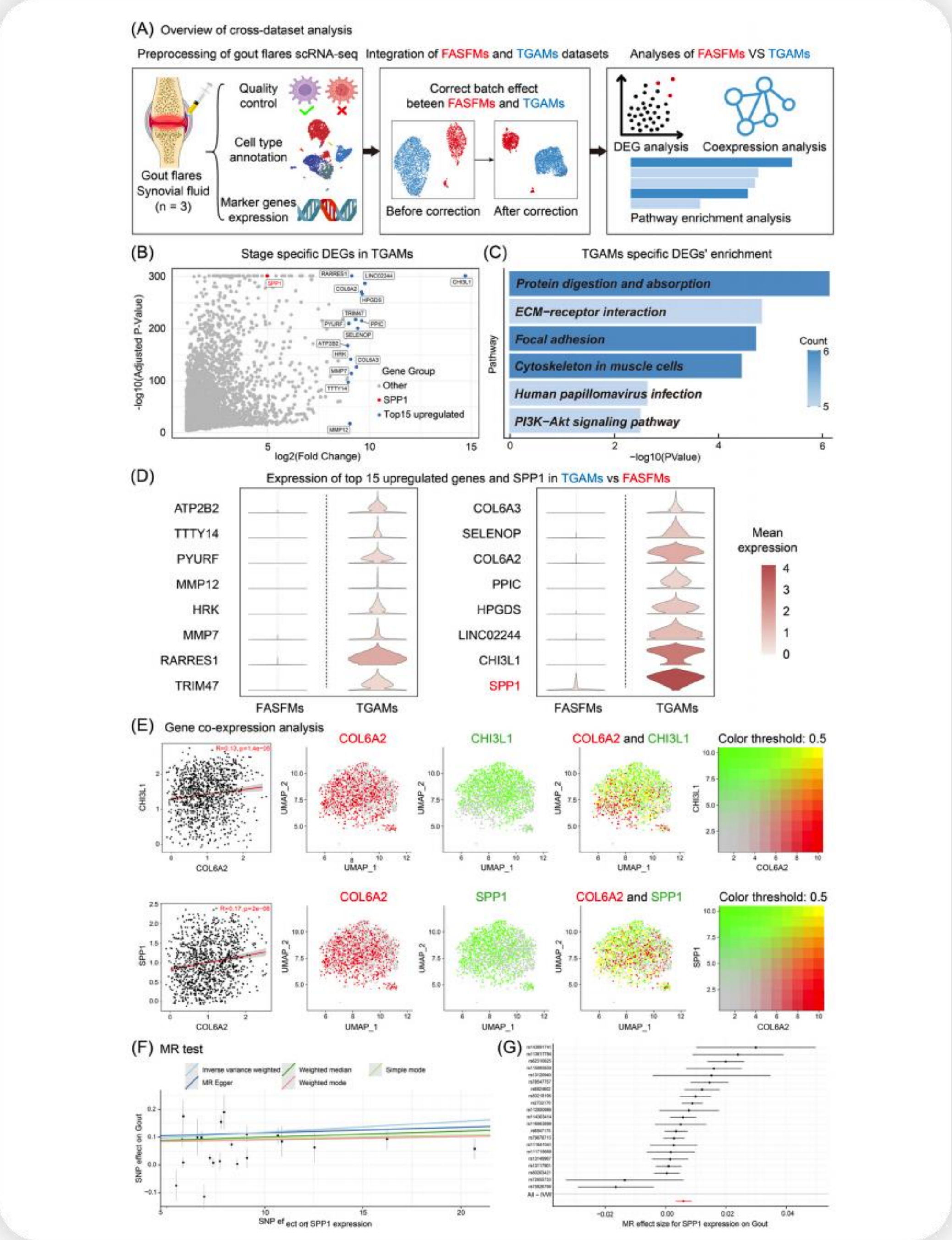
clusterProfiler

https://github.com/YuLab-SMU/clusterProfiler

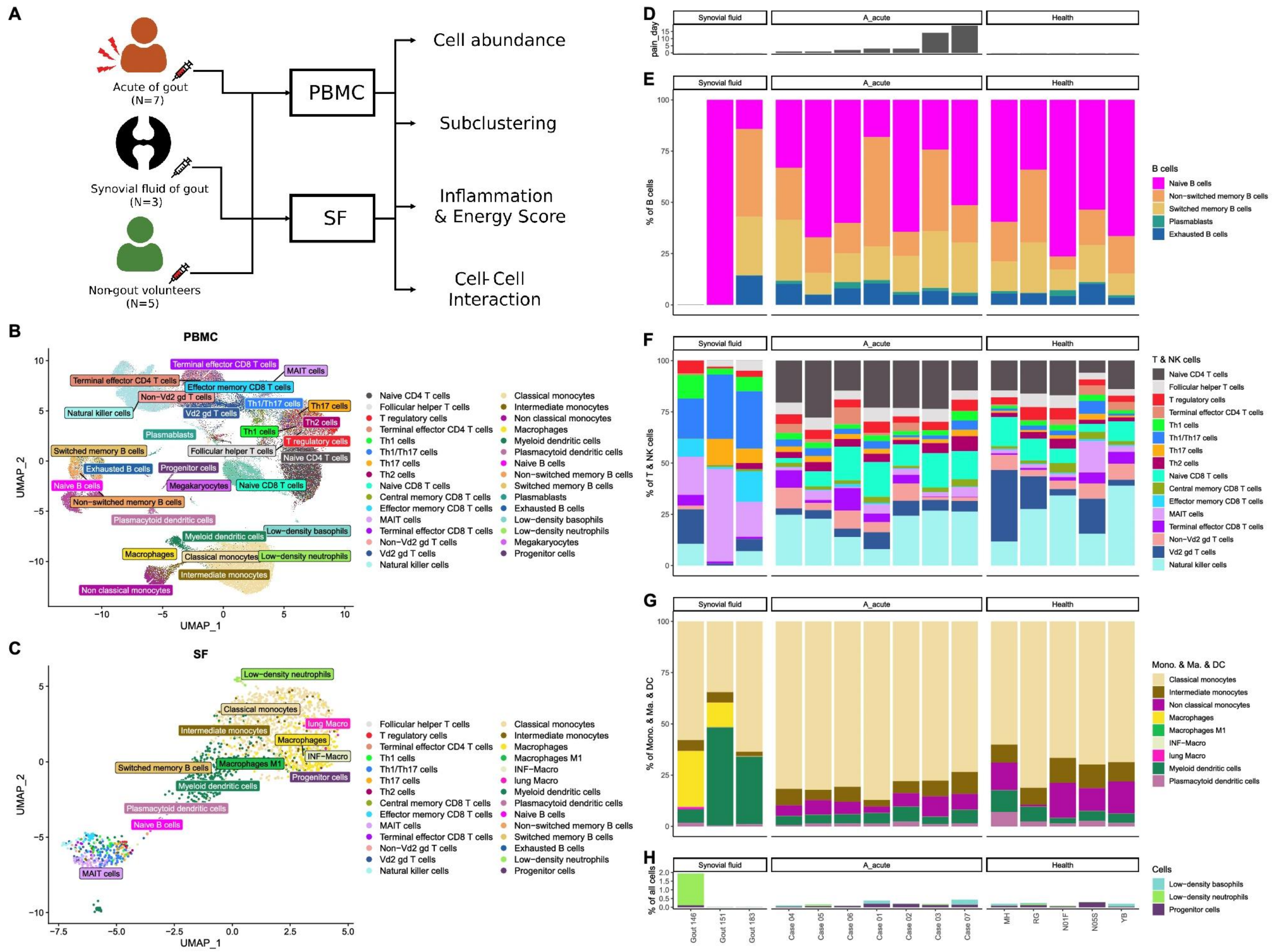
tophaceous gout-associated macrophages (TGAMs)痛风石性痛风相关巨噬细胞

intercritical gout-associated macrophages (IGAMs)间歇期痛风相关巨噬细胞

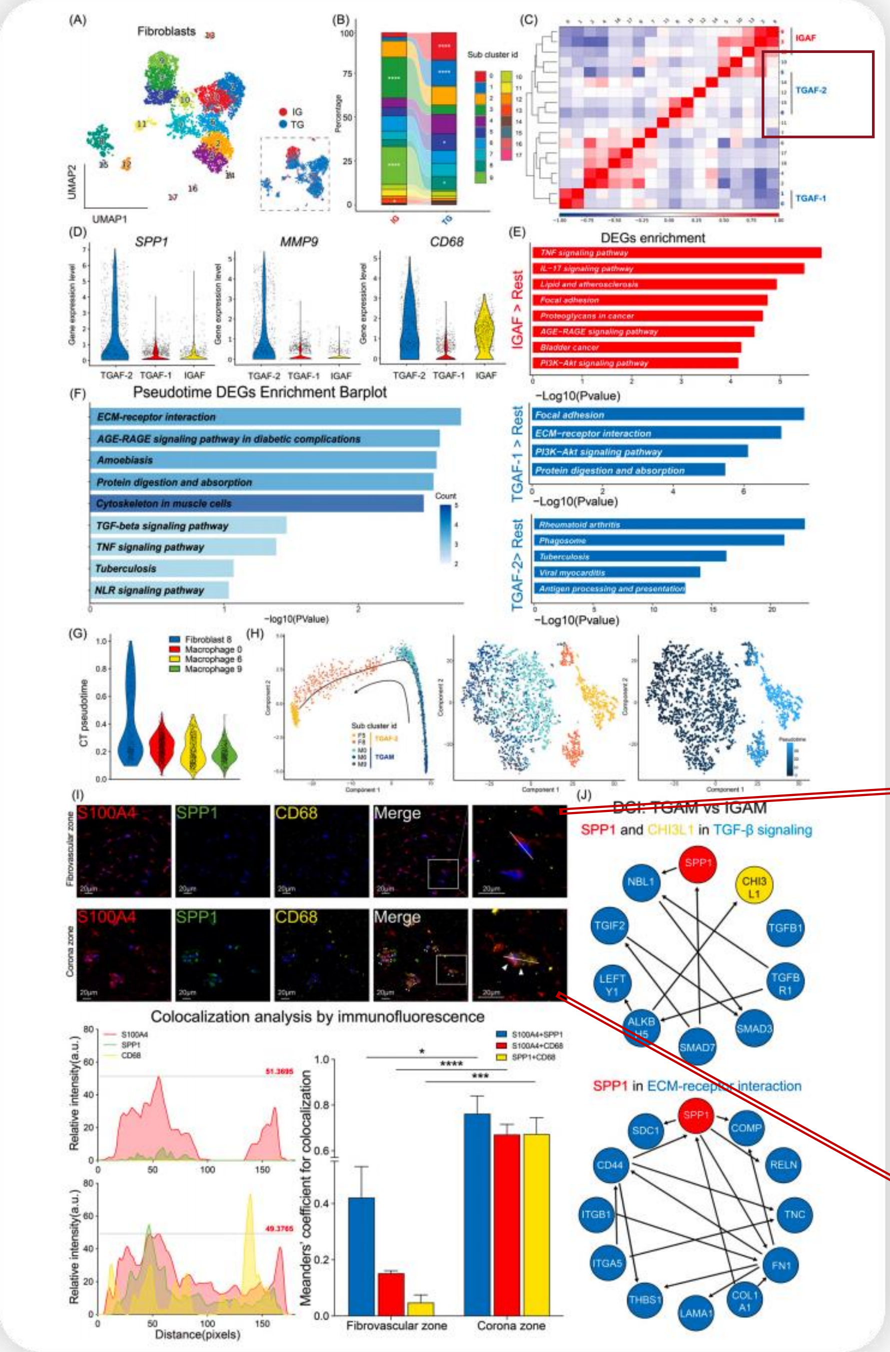




~~FASFMs 痛风急性发作相关滑液巨噬细胞~~



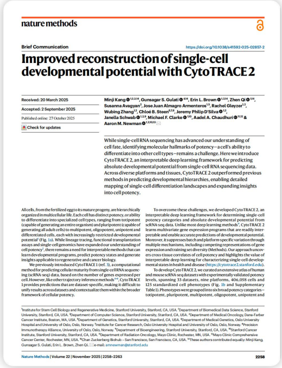
34 subpopulations (5 types of B cells, 16 types of T and natural killer cells, 9 types of monocytes, and 4 other cell types) in the blood of five healthy subjects and seven patients with acute gouty, and the SF of three patients with acute gout.



Python package for the creation, manipulation, and learning of Causal DAGs

inference causality causal-inference causal-models causal-dags

https://github.com/uhlerlab/causaldag



https://github.com/digitalcytometry/cytotrace2

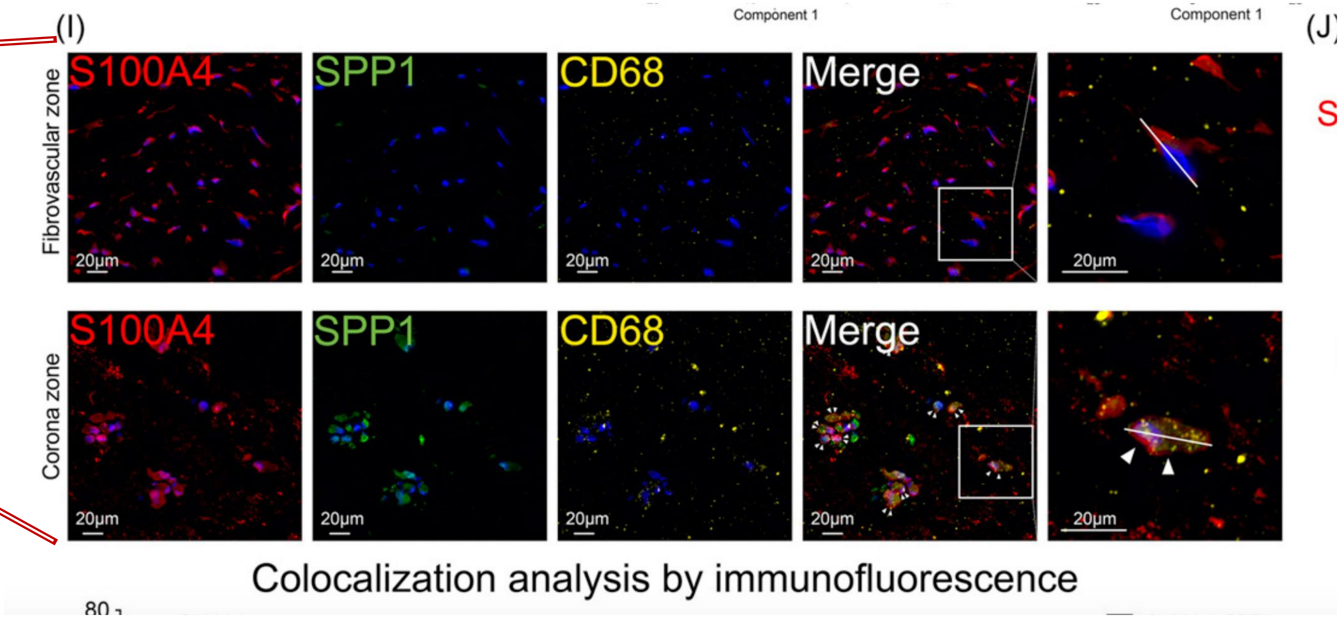
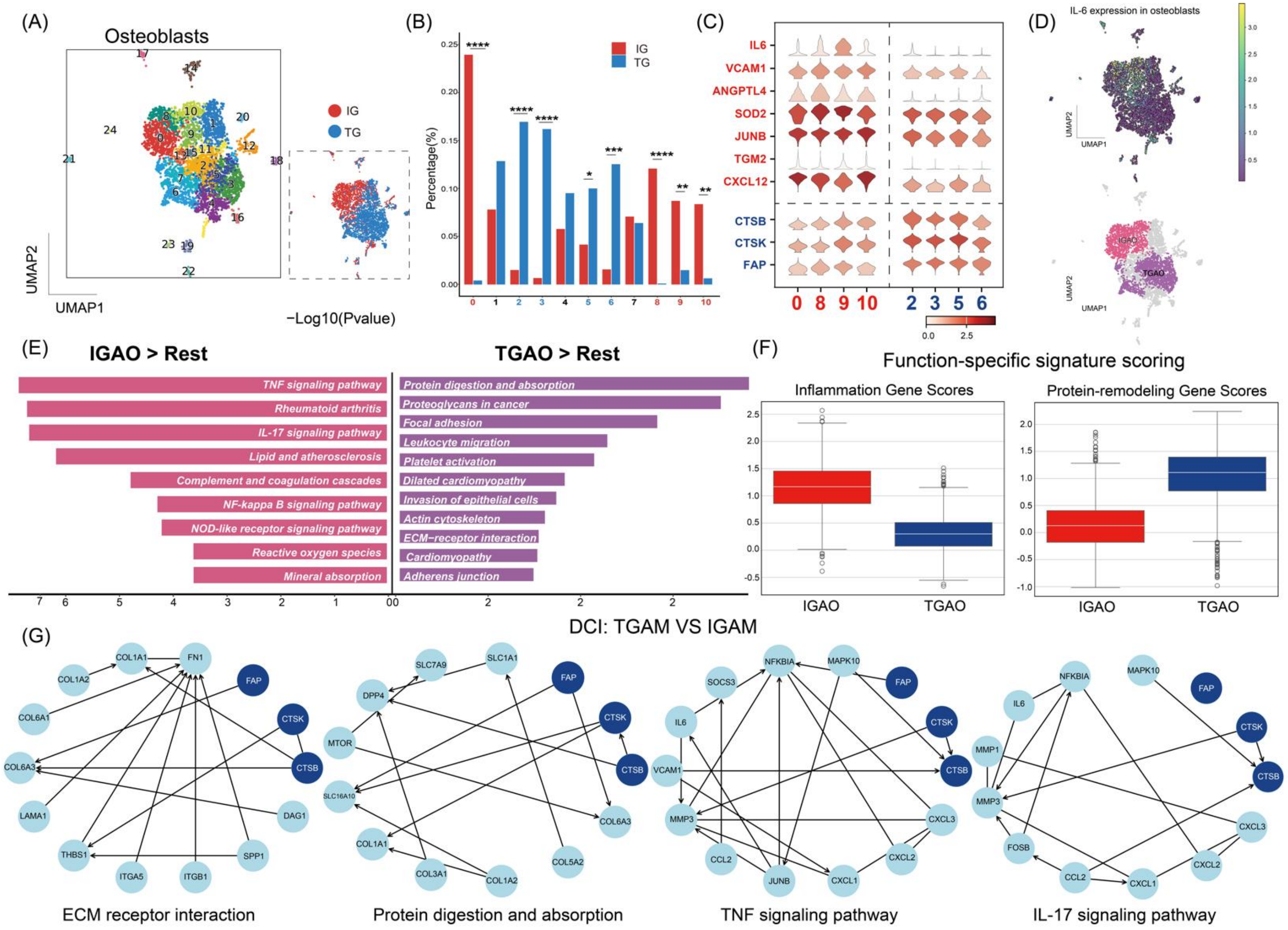
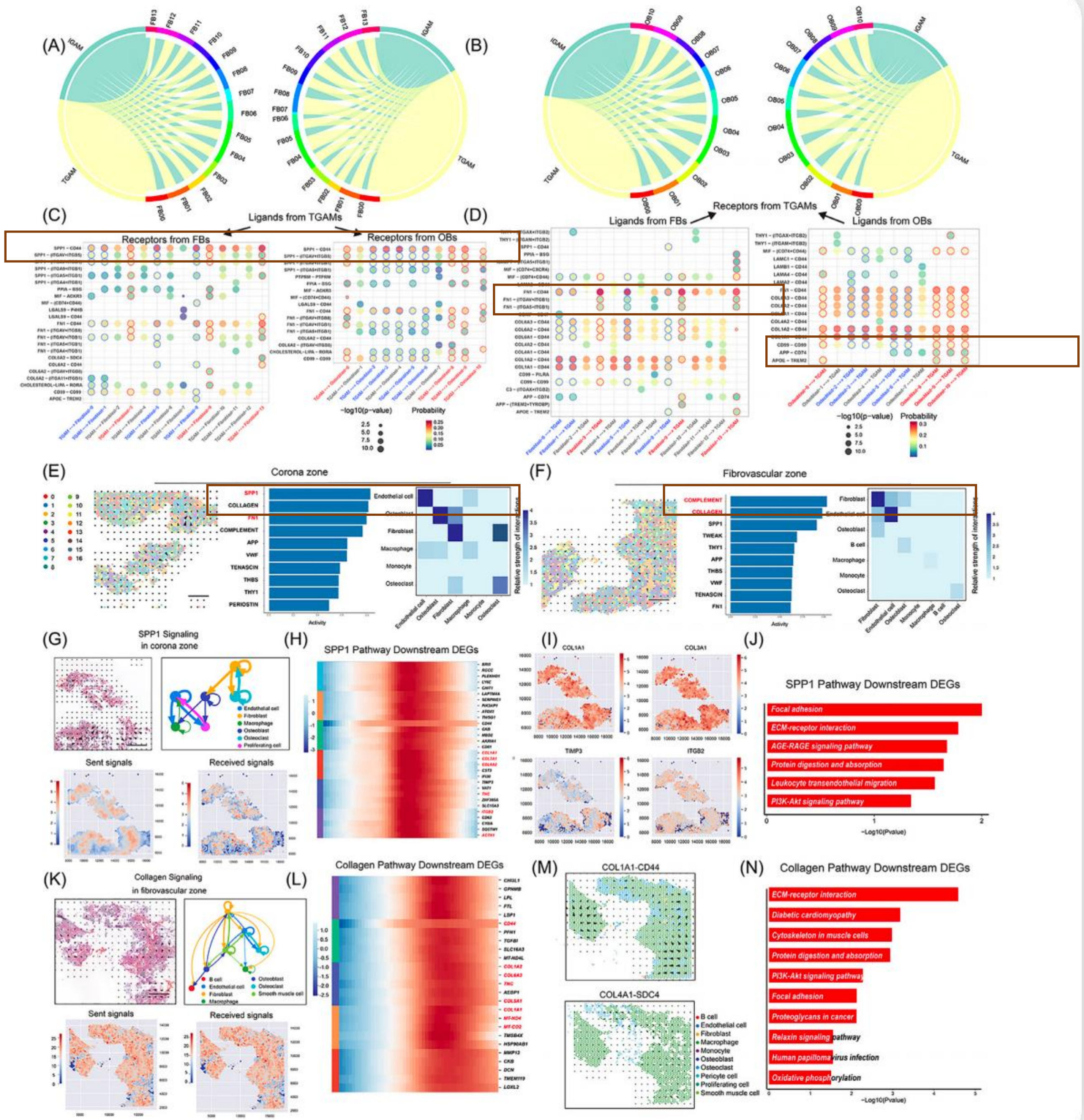


Figure S11



(A)
Osteoblasts
UMAP2
UMAP1
IG
TG
−Log10(Pvalue)
(B)
Percentage(%)
IG
TG
(C)
IL6
VCAM1
ANGPTL4
SOD2
JUNB
TGM2
CXCL12
CTSB
CTSK
FAP
0 8 9 10
2 3 5 6
(D)
IL-6 expression in osteoblasts
IGAO
TGAO
(E)
IGAO > Rest
TNF signaling pathway
Rheumatoid arthritis
IL-17 signaling pathway
Lipid and atherosclerosis
Complement and coagulation cascades
NF-kappa B signaling pathway
NOD-like receptor signaling pathway
Reactive oxygen species
Mineral absorption
TGAO > Rest
Protein digestion and absorption
Proteoglycans in cancer
Focal adhesion
Leukocyte migration
Platelet activation
Dilated cardiomyopathy
Invasion of epithelial cells
Actin cytoskeleton
ECM−receptor interaction
Cardiomyopathy
Adherens junction
(F)
Function-specific signature scoring
Inflammation Gene Scores
Protein-remodeling Gene Scores
IGAO
TGAO
DCI: TGAM VS IGAM
(G)
ECM receptor interaction
Protein digestion and absorption
TNF signaling pathway
IL-17 signaling pathway



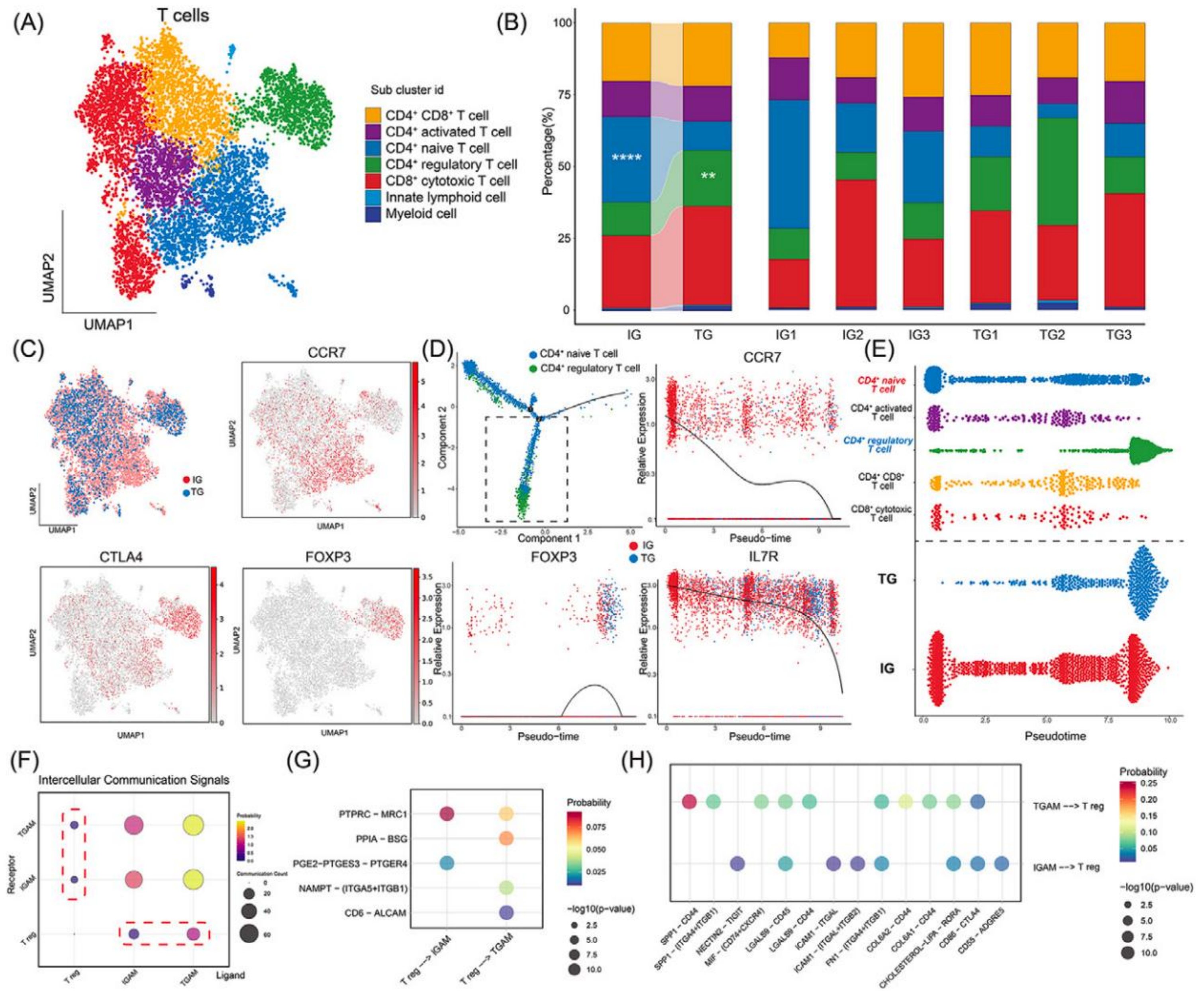
- Compared with intercritical gout, cell-cell communication is markedly enhanced in tophaceous gout.

- SPP1-CD44 is the dominant interaction linking SPP1+ TGAMs with fibroblasts and osteoblast-lineage cells.

- Spatially, SPP1 signalling predominates in the corona zone , whereas collagen signalling predominates in the fibrovascular zone

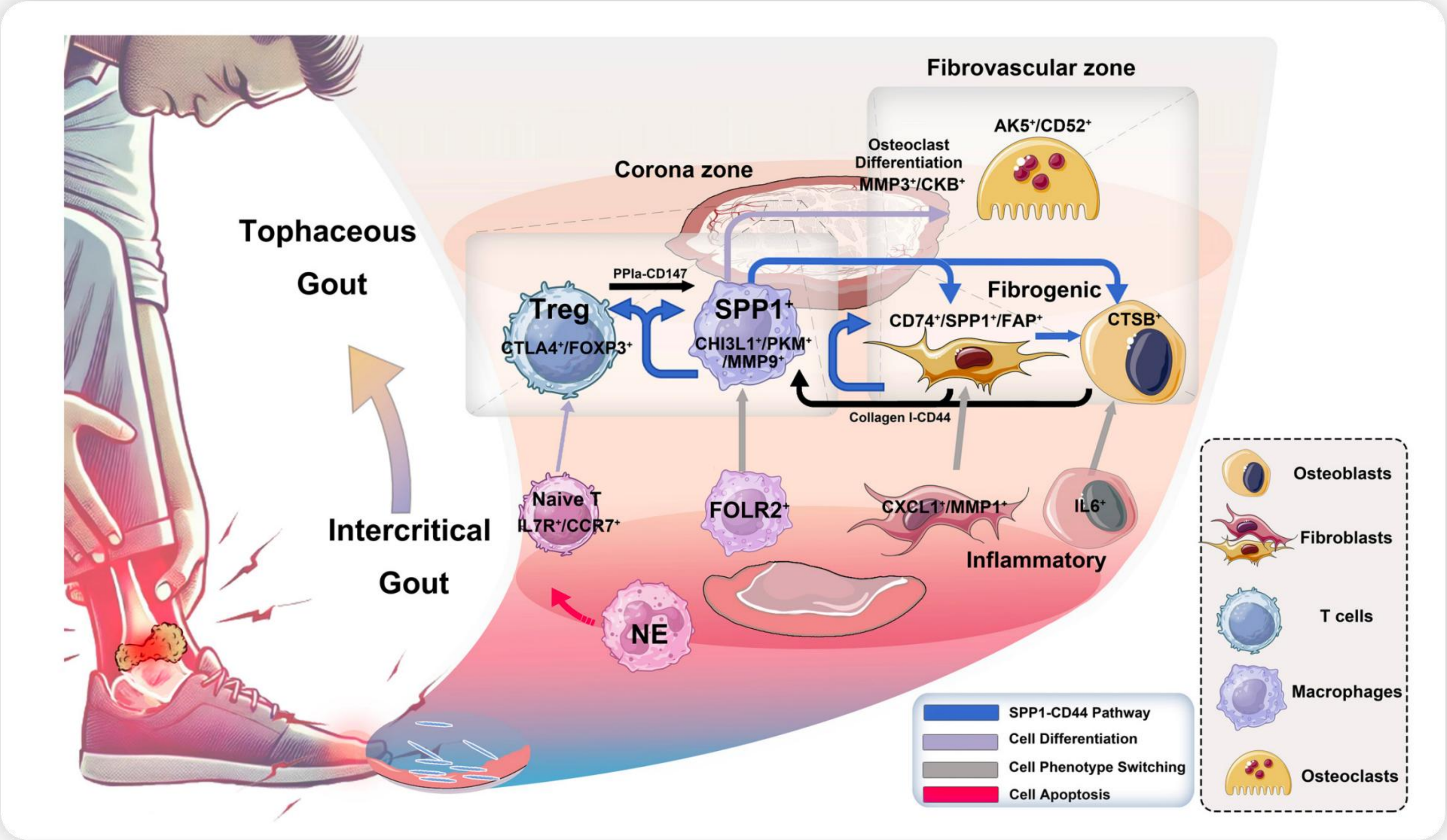
- These findings support a spatially organized communication network coupling chronic inflammation to ECM remodelling.

内层更偏 **SPP1** 驱动的免疫-基质互作，外层更偏 **collagen** 驱动的结构重塑



Celltype	GeneName	Celltype	GeneName
T_cell	CD3D	T_regulatory	FOXP3
	CD3E		CTLA4
	CD4		IL2RA
	CD8A	T_cytotoxic	GZMK
T_activated	CD69		GZMA
	TNF		NKG7
	FOS	ILC	KIT
	JUN		IL1R1
T_naive	CCR7		ID2
	SELL		AHR
	IL7R	Myeloid_cell	MMP9
	TCF7		LYZ

Supplemental table 3. List of T cells subclusters annotation markers.



Persistent hyperuricaemia

↓

MSU crystal deposition in joints

↓

Tophus microenvironment formation

↓

Expansion of SPP1+ TGAMs (SPP1+/MMP9+/CHI3L1+ macrophages)

↓

SPP1-CD44 signalling activation

↓

Crosstalk with fibroblasts, osteoblast/osteoclast-lineage cells, and Tregs

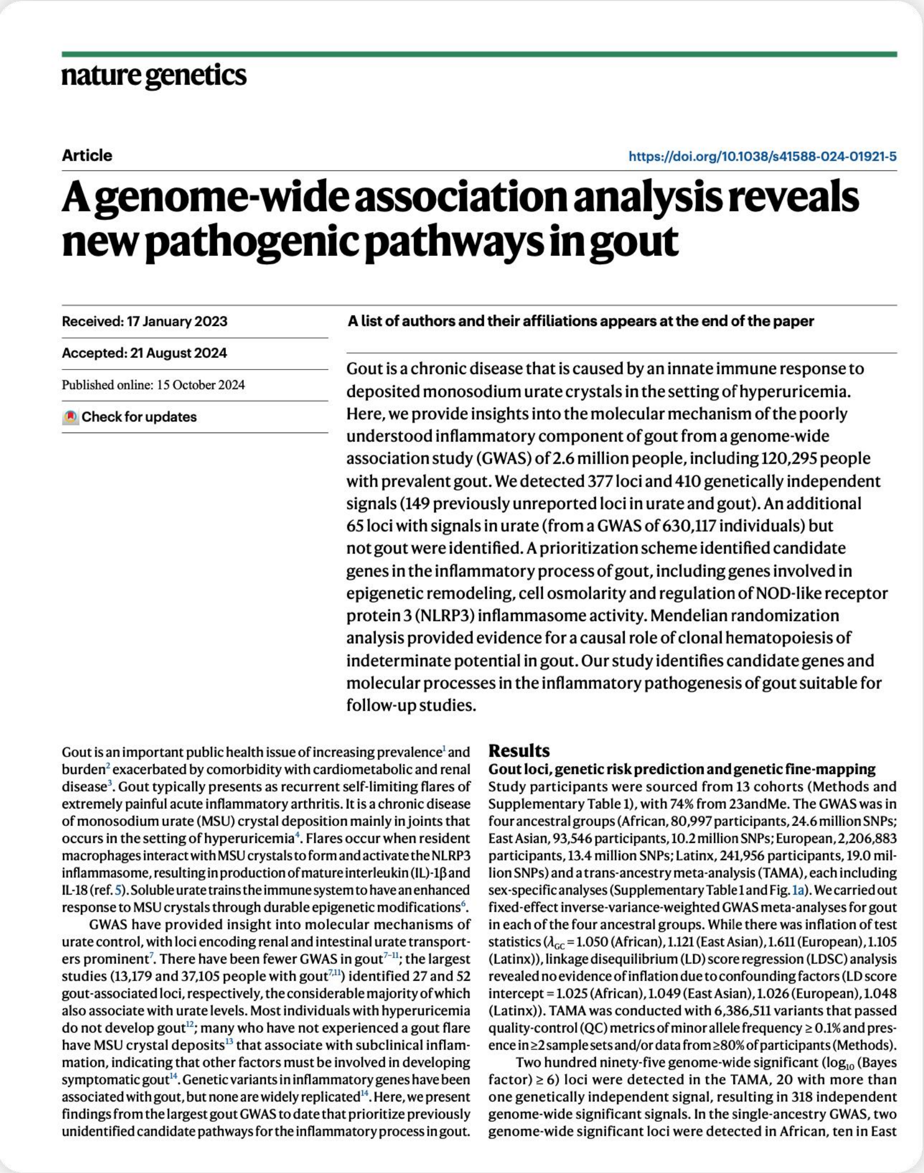
↓

ECM remodelling + fibrosis + osteoclast differentiation

↓

Tophus maturation and joint damage

Schematic of tophaceous gout patterns based on transcriptomics.


nature genetics

Article

https://doi.org/10.1038/s41588-024-01921-5

A genome-wide association analysis reveals new pathogenic pathways in gout

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A list of authors and their affiliations appears at the end of the paper

Gout is a chronic disease that is caused by an innate immune response to deposited monosodium urate crystals in the setting of hyperuricemia. Here, we provide insights into the molecular mechanism of the poorly understood inflammatory component of gout from a genome-wide association study (GWAS) of 2.6 million people, including 120,295 people with prevalent gout. We detected 377 loci and 410 genetically independent signals (149 previously unreported loci in urate and gout). An additional 65 loci with signals in urate (from a GWAS of 630,117 individuals) but not gout were identified. A prioritization scheme identified candidate genes in the inflammatory process of gout, including genes involved in epigenetic remodeling, cell osmolarity and regulation of NOD-like receptor protein 3 (NLRP3) inflammasome activity. Mendelian randomization analysis provided evidence for a causal role of clonal hematopoiesis of indeterminate potential in gout. Our study identifies candidate genes and molecular processes in the inflammatory pathogenesis of gout suitable for follow-up studies.

Gout is an important public health issue of increasing prevalence[1] and burden[2] exacerbated by comorbidity with cardiometabolic and renal disease[3]. Gout typically presents as recurrent self-limiting flares of extremely painful acute inflammatory arthritis. It is a chronic disease of monosodium urate (MSU) crystal deposition mainly in joints that occurs in the setting of hyperuricemia[4]. Flares occur when resident macrophages interact with MSU crystals to form and activate the NLRP3 inflammasome, resulting in production of mature interleukin (IL)-1β and IL-18 (ref. 5). Soluble urate trains the immune system to have an enhanced response to MSU crystals through durable epigenetic modifications[6].

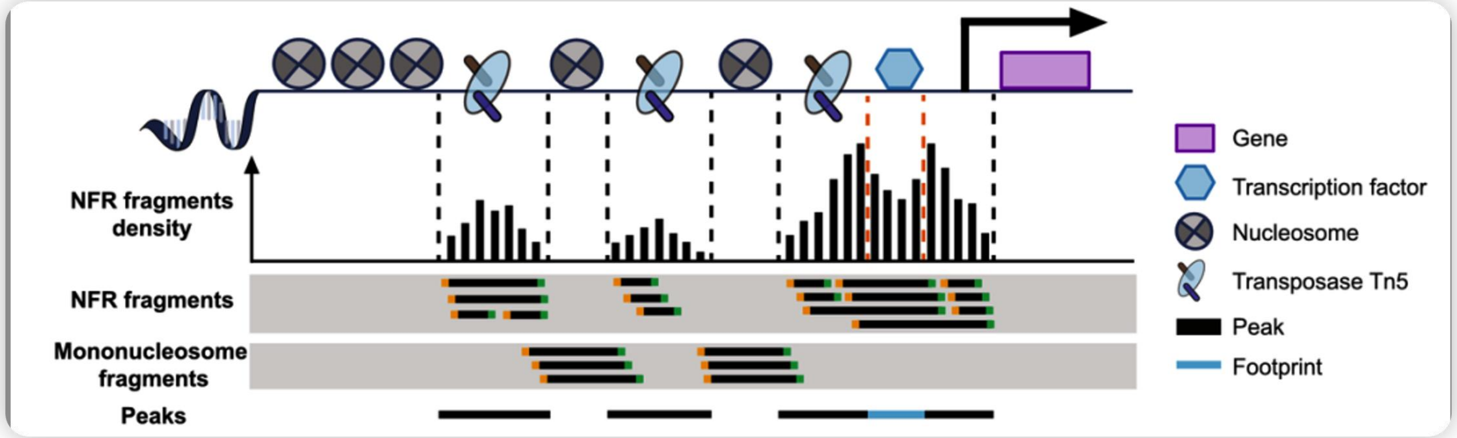
GWAS have provided insight into molecular mechanisms of urate control, with loci encoding renal and intestinal urate transporters prominent[7]. There have been fewer GWAS in gout[7–11]; the largest studies (13,179 and 37,105 people with gout[7,11]) identified 27 and 52 gout-associated loci, respectively, the considerable majority of which also associate with urate levels. Most individuals with hyperuricemia do not develop gout[12]; many who have not experienced a gout flare have MSU crystal deposits[13] that associate with subclinical inflammation, indicating that other factors must be involved in developing symptomatic gout[14]. Genetic variants in inflammatory genes have been associated with gout, but none are widely replicated[14]. Here, we present findings from the largest gout GWAS to date that prioritize previously unidentified candidate pathways for the inflammatory process in gout.

Results

Gout loci, genetic risk prediction and genetic fine-mapping

Study participants were sourced from 13 cohorts (Methods and Supplementary Table 1), with 74% from 23andMe. The GWAS was in four ancestral groups (African, 80,997 participants, 24.6 million SNPs; East Asian, 93,546 participants, 10.2 million SNPs; European, 2,206,883 participants, 13.4 million SNPs; Latinx, 241,956 participants, 19.0 million SNPs) and a trans-ancestry meta-analysis (TAMA), each including sex-specific analyses (Supplementary Table 1 and Fig. 1a). We carried out fixed-effect inverse-variance-weighted GWAS meta-analyses for gout in each of the four ancestral groups. While there was inflation of test statistics (λ_{GC} = 1.050 (African), 1.121 (East Asian), 1.611 (European), 1.105 (Latinx)), linkage disequilibrium (LD) score regression (LDSC) analysis revealed no evidence of inflation due to confounding factors (LD score intercept = 1.025 (African), 1.049 (East Asian), 1.026 (European), 1.048 (Latinx)). TAMA was conducted with 6,386,511 variants that passed quality-control (QC) metrics of minor allele frequency ≥ 0.1% and presence in ≥2 sample sets and/or data from ≥80% of participants (Methods).

Two hundred ninety-five genome-wide significant ($\log_{10}$ (Bayes factor) ≥ 6) loci were detected in the TAMA, 20 with more than one genetically independent signal, resulting in 318 independent genome-wide significant signals. In the single-ancestry GWAS, two genome-wide significant loci were detected in African, ten in East




Article

Spatially resolved mapping of cells associated with human complex traits

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Open access

Check for updates

Liyang Song[1,2,3], Wenhao Chen[1,2,3], Junren Hou[1,2], Minmin Guo[1,2] & Jian Yang[1,2]

Depicting spatial distributions of disease-relevant cells is crucial for understanding disease pathology[1,2]. Here we present genetically informed spatial mapping of cells for complex traits (gsMap), a method that integrates spatial transcriptomics data with summary statistics from genome-wide association studies to map cells to human complex traits, including diseases, in a spatially resolved manner. Using embryonic spatial transcriptomics datasets covering 25 organs, we benchmarked gsMap through simulation and by corroborating known trait-associated cells or regions in various organs. Applying gsMap to brain spatial transcriptomics data, we reveal that the spatial distribution of glutamatergic neurons associated with schizophrenia more closely resembles that for cognitive traits than that for mood traits such as depression. The schizophrenia-associated glutamatergic neurons were distributed near the dorsal hippocampus, with upregulated expression of calcium signalling and regulation genes, whereas depression-associated glutamatergic neurons were distributed near the deep medial prefrontal cortex, with upregulated expression of neuroplasticity and psychiatric drug target genes. Our study provides a method for spatially resolved mapping of trait-associated cells and demonstrates the gain of biological insights (such as the spatial distribution of trait-relevant cells and related signature genes) through these maps.

The composition and spatial organization of cells within a tissue are critical to its function and can also serve as indicators of its health status[1]. The advancement in spatial transcriptomics (ST), which enables the profiling of gene expression levels of cells in their native spatial positions, represents a promising avenue for investigating cell spatial organizations and uncovering related biological mechanisms[3,4]. In recent years, an increasing number of studies have utilized ST technologies to explore cell spatial organization in diverse tissues[5–7]. However, a substantial knowledge gap persists in identifying cells that are most relevant to complex traits or diseases and mapping their spatial distribution within a tissue.

To identify trait-associated cells or cell types, previous studies have proposed genetics-informed strategies that integrate data from genome-wide association studies (GWAS) of complex traits, including diseases, with single-cell RNA-sequencing (scRNA-seq) data[8–11]. Although these methods can pinpoint trait-associated cells, they encounter challenges in mapping the spatial distribution of these identified cells owing to the lack of cell spatial positional information in scRNA-seq data. These scRNA-seq-based methods can, in principle, be applied to ST data. However, owing to the absence of modelling for cell spatial coordinates and the high level of technical noise in ST data[12], they have limited power in spatially aware mapping of trait-associated cells. Although previous studies[13] have associated spatial domains with complex traits, these analyses are not at single-cell resolution, limiting their ability to profile the spatial distribution of trait-associated cells. Consequently, there is a need for new methods that can integrate ST data into GWAS for fine-scale spatially resolved mapping of trait-associated cells.

In this study, we introduce gsMap, a method that integrates high-resolution ST data and GWAS summary statistics for spatially resolved mapping of trait-associated cells. Utilizing embryonic ST datasets covering 25 organs, we assessed the specificity of gsMap through simulated GWAS data and the sensitivity of the method by recapitulating known associations between cells in different organs and a range of complex traits. Applying gsMap to brain ST datasets, we generated extensive trait–brain cell association maps encompassing 30 human brain-related complex traits.

Overview of gsMap

The fundamental concept of gsMap involves assessing whether genetic variants, predominantly single nucleotide polymorphisms (SNPs), located in or near genes highly expressed in a spot in ST data are enriched for genetic associations with a trait of interest (Methods and Supplementary Note, sections 1–3). Each ST section is required to include transcriptome-wide gene expression profiles and spatial coordinates of individual spots. Here, a 'spot' refers to a cell in high-resolution ST platforms (for example, Stereo-seq cell-bin mode) or a cluster of cells in conventional ST platforms (such as 10X Visium). gsMap consists of three steps. First, gsMap leverages spot homogeneity to address sparsity and technical noise in ST data. gsMap uses a graph neural network (GNN) to identify homogeneous spots for each focal spot in terms of

[1]School of Life Sciences, Westlake University, Hangzhou, China. [2]Westlake Laboratory of Life Sciences and Biomedicine, Hangzhou, China. [3]These authors contributed equally: Liyang Song, Wenhao Chen. ✉e-mail: jian.yang@westlake.edu.cn

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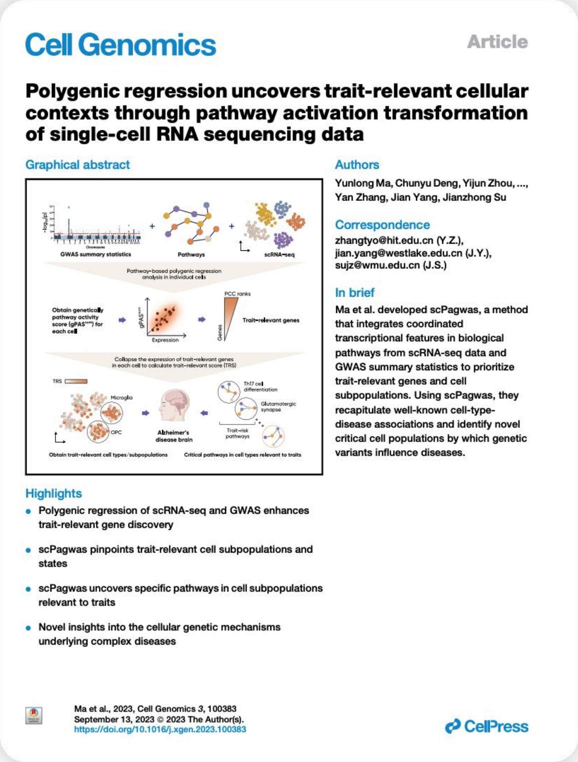
Cell Genomics

Article

Polygenic regression uncovers trait-relevant cellular contexts through pathway activation transformation of single-cell RNA sequencing data

Graphical abstract

GWAS summary statistics
Pathways
scRNA-seq
Pathway-based polygenic regression analysis in individual cells
Obtain genetically pathway activity score (gPAS) for each cell
Expression
PCC ranks
Genes
Trait-relevant genes
Collapse the expression of trait-relevant genes in each cell to calculate trait-relevant score (TRS)
TRS
Microglia
OPC
Alzheimer's disease brain
Th17 cell differentiation
Glutamatergic synapse
Trait-risk pathways
Obtain trait-relevant cell types/subpopulations
Critical pathways in cell types relevant to traits

Highlights

- Polygenic regression of scRNA-seq and GWAS enhances trait-relevant gene discovery
- scPagwas pinpoints trait-relevant cell subpopulations and states
- scPagwas uncovers specific pathways in cell subpopulations relevant to traits
- Novel insights into the cellular genetic mechanisms underlying complex diseases

Authors

Yunlong Ma, Chunyu Deng, Yijun Zhou, ..., Yan Zhang, Jian Yang, Jianzhong Su

Correspondence

zhangtyo@hit.edu.cn (Y.Z.), jian.yang@westlake.edu.cn (J.Y.), sujz@wmu.edu.cn (J.S.)

In brief

Ma et al. developed scPagwas, a method that integrates coordinated transcriptional features in biological pathways from scRNA-seq data and GWAS summary statistics to prioritize trait-relevant genes and cell subpopulations. Using scPagwas, they recapitulate well-known cell-type-disease associations and identify novel critical cell populations by which genetic variants influence diseases.

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